Jonathan Jamison

Software Engineer | Modelling & Simulation

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Versatile engineer with a future-focused skillset and a passion for programming, performance, and emerging technologies. Delivered award-winning work across academia and fast-paced startups through the development of novel solutions to complex engineering problems. Continuously deepening expertise through certifications, personal projects, and postgraduate study across multiple domains.

Core Competencies

- Application & Toolset Development
- Dynamic Modelling & Simulation
- Discrete Event Modelling
- Data Analysis & Visualisation
- Machine Learning & Data Pipelines
- Automation & Optimisation
- System Design
- Research & Development
- Technical Consultation

Career Experience

Kallikor, London, United Kingdom — March 2026 – Present

Software Engineer, Modelling

Software Engineer at a defence-born simulation startup, building digital twin models that help commercial leaders stress-test decisions and optimise complex operational systems.

- Delivering simulation and analysis projects for multi-billion-pound enterprise customers, combining software engineering, data analysis, and technical consultancy to identify operational improvements.
- Developing scalable Python-based modelling frameworks, algorithms and simulation tooling to represent real-world operations, networks, and autonomous processes.
- Building pipelines to transform multi-domain customer data into representative simulation flows, modelling real-world processes and communicating findings to support evidence-led decision-making.
- Collaborating across engineering and product teams to design maintainable architectures, improve simulation performance, and integrate modern software engineering practices.

ClearMotion, London, United Kingdom — August 2024 – March 2026

Modelling & Simulation Engineer

Senior Engineer at an automotive-tech startup driving innovation in motion-control by integrating machine learning with vehicle dynamics and high-performance hardware.

- Developed, deployed, and continuously improved bespoke simulation and analysis tools for internal and external customers, applying modern programming practices and integrating new technologies.
- Lead engineer on ‘Virtual Sensor’ project to replace physical component with a machine learning model. Developed an automated workflow to perform model training, evaluation, containerisation, analysis and optimisation.
- Worked with large-scale vehicle telemetry datasets, supporting data acquisition, workflows, and data analysis techniques to communicate insights to Director-level stakeholders.
- Applied vehicle dynamics and multi-domain modelling techniques to develop and validate high-fidelity 14-DoF vehicle model to support system performance and robustness targets.

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Rivian Automotive, Woking, United Kingdom February 2022 – August 2024
Vehicle Modelling Engineer

Member of a cross-functional simulation team pioneering a software-led approach to vehicle development at a globally renowned electric vehicle startup; recipient of *Top Talent Award* for outstanding performance within role.

- Contributed to the creation and correlation of full-vehicle digital twins, developing subsystem models, test suites, optimisation workflows, and simulation infrastructure to support product delivery across multiple vehicle attributes.
- Developed simulation tooling for optimisation, automation, and multi-domain analysis, improving the efficiency and repeatability of vehicle development workflows.
- Supported reduction of physical testing requirements by assisting Driver-in-the-Loop sessions through experiment design, live coding, model integration, automation routines, and operational support.

IPG Automotive & Ford UK | Graduate Application & Simulation Engineer February 2021 – February 2022
Resident Application Engineer supporting industry-leading vehicle simulation projects.

Generac Power Systems | Engineering Intern, Powertrain Development Summer 2018 & 2019
Successfully completed two summer internships at Generac HQ in Wisconsin, USA.

Education

Postgraduate Certificate in Artificial Intelligence (PGCert), *with Distinction*
Ulster University, Belfast, Northern Ireland
Masters-level modules studied as part of scholarship, including Big Data & Infrastructure, Machine Learning, Statistical Modelling & Data Mining.

Master of Engineering in Mechanical Engineering (MEng), *1st Class with Honours, Degree+*
Queen's University Belfast, Northern Ireland
Award winning final year project centred around the creation of a high-fidelity vehicle dynamics model, inclusive of machine learning for optimal parameter generation. STEM Ambassador & Peer Assisted Learning mentor.
Queen's Formula Racing Suspension Team Leader & Assistant Performance Team Leader.

Honours & Awards

Global Undergraduate Awards Winner & Highly Commended Entrant *for final year project, 2020*

Outstanding Project Award (2nd Place) *for final year project, presented by IMechE & NAFEMS, 2020*

Category Winner and Overall Finalist, *What's the Big Idea? & QUBSU Dragons Den, SU Enterprise, 2019*

Finalist, Leaders within Engineering Scholarship, *Royal Academy of Engineering, 2018*

Personal Projects

Spring-Mass-Damper Machine Learning Model |Python, Plotly Dash, FastAPI

Interactive web app to replicate the dynamics of a mechanical system using a trained neural-network surrogate.

Bicycle Model Neural Network | MATLAB, Simulink, Python, TensorFlow

Replicating behaviour of non-linear system using automated workflow to simulate vehicle dynamics.

Technical Proficiencies

Python, NumPy, pandas, SciPy, TensorFlow, MATLAB, Simulink, FastAPI, Docker, Git, AWS, IPG CarMaker, Jira; currently learning C++ and SQL.